

Perfboard Wiring Sheet — Treadmill HAT

Board: plain perfboard, 24 cols × 20 rows. Holes referenced as (row, col).

Layout: - Pi GPIO socket: pins 1-40 in rows 3-4, cols 3-22 - +8V bus (bare wire): row 6, cols 2-22 - GND bus (bare wire): row 8, cols 2-22 - Components (R1, R2, R3, U1, C3, C4): rows 10-12 - J1 (Console) 2×4 IDC header: rows 17-18, cols 2-5 (odd pins on row 18 / near edge, even pins on row 17) - J2 (Motor) 2×4 IDC header: rows 17-18, cols 13-16 (odd pins on row 18 / near edge, even pins on row 17)

Grid layout

	01	02	03	04	05	06	07	08	09	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24
R01	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
R02	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
R03	.	.	01	03	05	07	09	11	13	15	17	19	21	23	25	27	29	31	33	35	37	39	.	.
R04	.	.	02	04	06	08	10	12	14	16	18	20	22	24	26	28	30	32	34	36	38	40	.	.
R05	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
R06	.	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	.	.
R07	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
R08	.	G	G	G	G	G	G	G	G	G	G	G	G	G	G	G	G	G	G	G	G	G	.	.
R09	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
R10	.	.	.	.	.	.	.	.	.	.	.	.	Vo	Gu	Vi	Sd	Pg	.	.	.	.	.	.	.
R11	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
R12	.	.	.	.	.	R1	.	.	.	R1	.	.	.	.	.	.	.	.	.	.	.	.	.	.
R13	.	.	.	.	.	R2	.	.	.	R2	.	.	.	.	.	.	.	.	.	.	.	.	.	.
R14	.	.	.	.	.	R3	.	.	.	R3	.	.	.	.	.	.	.	.	.	.	.	.	.	.
R15	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
R16	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
R17	.	2	4	6	8	.	.	.	.	.	.	.	2	4	6	8	.	.	.	.	.	.	.	.
R18	.	1	3	5	7	.	.	.	.	.	.	.	1	3	5	7	.	.	.	.	.	.	.	.
R19	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
R20	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.

Legend: - . — empty perfboard hole - 01–40 — Pi GPIO pin (row 3 = odd pins, row 4 = even pins). Pi sits in 2×20 socket at cols 3–22. - + — +8V bus (bare wire across row 6, cols 2–22) - G — GND bus (bare wire across row 8, cols 2–22) - Vo — U1 pin 1 VOUT (+5V output) - Gu — U1 pin 2 GND - Vi — U1 pin 3 VIN (+8V input) - Sd — U1 pin 4 ~SHDN (leave floating) - Pg — U1 pin 5 PG (leave floating) - R1 / R2 / R3 — resistor leads at cols 6 and 10 (body spans cols 7–9 over the board) - Numbers 1–8 at rows 17–18 — RJ45 pin via 2×4 IDC ribbon header. Standard IDC numbering: **odd pins on row 18 (near edge), even pins on row 17 (far edge)**. Left block (cols 2–5) is **J1 (Console)**, right block (cols 13–16) is **J2 (Motor)**. - C3 leads share holes (10,14) and (10,15) with U1 GND/VIN. C4 leads share (10,13) and (10,14) with U1 VOUT/GND. Solder caps directly across the U1 pins on top side; U1 pins go through the same holes.

Component placements

Component	Position	Notes
U1 D24V10F5 (5 pins horizontal)	row 10, cols 13-17	Pin 1 (VOUT) at (10, 13), Pin 5 (PG) at (10, 17)
C3 ceramic disc 100nF	leads share holes with U1	Lead 1 in (10, 14) [U1 GND]; Lead 2 in (10, 15) [U1 VIN]
C4 ceramic disc 100nF	leads share holes with U1	Lead 1 in (10, 13) [U1 VOUT]; Lead 2 in (10, 14) [U1 GND]
R1 100Ω horizontal	leads at (12, 6) and (12, 10)	Motor read line — left lead → J1/J2 pin 3, right lead → Pi GP17
R2 100Ω horizontal	leads at (13, 6) and (13, 10)	Console read — left → J1 pin 6, right → Pi GP27
R3 100Ω horizontal	leads at (14, 6) and (14, 10)	Motor write — left → J2 pin 6, right → Pi GP22

(R1/R2/R3 stacked tightly at cols 6 and 10. Adjust spacing if leads touch.)

Hole → Pi GPIO pin reference

Pi pin N is at: **odd N** → **row 3, col 3 + (N-1)/2**; **even N** → **row 4, col 3 + (N-2)/2**

Key pins you'll use:

Pi Pin	Function	Hole
2	+5V	(4, 3)
6	GND	(4, 5)
11	GP17	(3, 8)
13	GP27	(3, 9)
15	GP22	(3, 10)

Jumper list (back of board, insulated wire)

Power rails (most critical — do first)

#	From hole	To hole	Wire
1	J1 pin 1 (18, 2)	GND bus (8, 2)	thin (long, row 18 → row 8)
2	J1 pin 7 (18, 5)	GND bus (8, 5)	thin (long, row 18 → row 8)
3	J2 pin 1 (18, 13)	GND bus (8, 13)	thin (long)
4	J2 pin 7 (18, 16)	GND bus (8, 16)	thin (long)
5	U1 GND (10, 14)	GND bus (8, 14)	thin
6	Pi pin 6 GND (4, 5)	GND bus (8, 5)	thin (shares hole with #2 at the bus end)
7	J1 pin 2 (17, 2)	+8V bus (6, 2)	thin (insulated, hops over GND bus)
8	J1 pin 8 (17, 5)	+8V bus (6, 5)	thin (insulated, hops over GND bus)
9	J2 pin 2 (17, 13)	+8V bus (6, 13)	thin (insulated, hops over GND bus)
10	J2 pin 8 (17, 16)	+8V bus (6, 16)	thin (insulated, hops over GND bus)
11	U1 VIN (10, 15)	+8V bus (6, 15)	thin

+5V from regulator to Pi

#	From	To	Wire
12	U1 VOUT (10, 13)	Pi pin 2 +5V (4, 3)	thin

Pass-through (J1 ↔ J2 direct wires)

#	From	To	Wire
13	J1 pin 4 (17, 3)	J2 pin 4 (17, 14)	thin (long, spans 11 cols, runs along row 17)
14	J1 pin 5 (18, 4)	J2 pin 5 (18, 15)	thin (long, spans 11 cols, runs along row 18)

Motor read tap (pin 3, both jacks share, Pi listens)

#	From	To	Wire
15	J1 pin 3 (18, 3)	J2 pin 3 (18, 14)	thin (long, spans 11 cols, runs along row 18 — parallel to #14, keep them apart)
16	J1 pin 3 (18, 3)	R1 left lead (12, 6)	thin
17	R1 right lead (12, 10)	Pi GP17 pin 11 (3, 8)	thin (will hop over buses — use insulated wire)

Console read (Pi reads from J1 side of pin 6)

#	From	To	Wire
18	J1 pin 6 (17, 4)	R2 left lead (13, 6)	thin
19	R2 right lead (13, 10)	Pi GP27 pin 13 (3, 9)	thin (hops over buses)

Motor write (Pi writes to J2 side of pin 6)

#	From	To	Wire
20	J2 pin 6 (17, 15)	R3 left lead (14, 6)	thin (long, spans 9 cols)
21	R3 right lead (14, 10)	Pi GP22 pin 15 (3, 10)	thin (hops over buses)

Total: 21 jumper wires + 2 power buses

Smoke test (BEFORE inserting Pi, BEFORE plugging in cable)

- ☐ Multimeter beep mode: GND bus (row 8) is one continuous net
- ☐ Multimeter: +8V bus (row 6) is one continuous net
- ☐ NO continuity between +8V bus and GND bus (no short!)
- ☐ NO continuity between +5V trace (10, 13) and any other rail
- ☐ U1 oriented correctly: pin 1 VOUT is leftmost (col 13), pin 5 PG is rightmost (col 17)
- ☐ Pi NOT inserted yet
- ☐ J1 and J2 headers seated firmly

First power test

- ☐ Plug treadmill cable into J2 only (the motor side, since that's where +8V comes from)
- ☐ Multimeter on +8V bus reads ~8V (acceptable: 7-9V)
- ☐ Multimeter on U1 VOUT (10, 13) reads ~5V (acceptable: 4.9-5.1V)
- ☐ NOW insert Pi

Pi software setup

- ☐ Pi boots (PWR LED solid, ACT LED blinks)
- ☐ Configure GPIOs: GP17 = INPUT, GP27 = INPUT, GP22 = OUTPUT
- ☐ Test reading GP17 — should see motor data when console is also connected

⚠ DO NOT

- ❌ Do NOT wire J1 pin 6 to J2 pin 6 (they MUST be separate — Pi is in the middle)
 - ❌ Do NOT wire U1 VOUT to Pi pin 1 or pin 17 (those are +3.3V — would destroy Pi)
 - ❌ Do NOT power up the Pi until you've verified +5V is correct on U1 output
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Bus jumpers cross-over note

Use **insulated wire on the back of the board for ALL jumpers** — too many cross over the bare bus wires:
- #7, #8, #9, #10 (J1/J2 +8V wires) cross OVER the GND bus on the way from row 17 up to row 6. - #17, #19, #21 (resistor right leads → Pi GPIO) cross OVER both the +8V bus (row 6) and GND bus (row 8) on the way from rows 12–14 up to row 3. - #14 and #15 (pin 5 pass-through and pin 3 tap) both run parallel along row 18 for ~11 cols. Keep them physically separated or use a slight offset — they're adjacent (row 18, cols 3–4 to cols 14–15).

Bare wire on any of these = short. Insulated wire (24 AWG hookup) only.